

Fig. 8: The GRIN structure (Courtesy: NHK Science and Technology Research Laboratories)

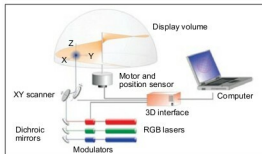


Fig. 9: A volumetric display functioning (Courtesy: www.google.com)

image is produced in a physical space rather than on a flat screen. The image

is said to contain voxels, that is, volumetric pixels that have x , y and depth z coordinated. The voxels appear at different physical depths within the space, thus allowing the eye to converge and focus just as they would converge in the real world. The first volumetric display was created by W.D. Chase in 1976. He used a rotating helix screen.

A volumetric display comprises a rotating helical screen (semi-transparent), computer interface and laser system. The helical screen rotates at such a high speed that it becomes invisible to eyes. The computer provides the required control signals (on/off durations and intensity control) to the 3D interface, which then modulates the required colour combination.

As the light beam hits the rotating screen, it scatters and causes a voxel.

The location of this voxel is determined by the intersection of light beam with the rotating helix. Expanding this concept further, instead of using helix structure, an LED array screen is rotated at high speed.

Health-related side effects

Watching excessive 3D content can be very harmful not only for the eyes but also for the brain. The projected 3D content is very much different from the real world that our eyes see. Chromatic adaptation of individuals also plays an important role in the harmful effects of 3D. Some common effects include:

1. Motion sickness
2. Excessive eye strain
3. Disturbance of postural balance that leads to disorientation problems
4. Increase in heart beat
5. Headache and increased stress levels

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